Constructions With Monoidal Categories

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November 22, 2024

01UE This chapter contains some material on constructions with monoidal categories.

Contents

1 Moduli Categories of Monoidal Structures **1**
1.1 The Moduli Category of Monoidal Structures on a Category 1
1.2 The Moduli Category of Braided Monoidal Structures on a Category 9
1.3 The Moduli Category of Symmetric Monoidal Structures on a Category 9

2 Moduli Categories of Closed Monoidal Structures **9**

3 Moduli Categories of Refinements of Monoidal Structures **9**
3.1 The Moduli Category of Braided Refinements of a Monoidal Structure 9

A Other Chapters **10**

01UF 1 Moduli Categories of Monoidal Structures

01UG 1.1 The Moduli Category of Monoidal Structures on a Category

Let C be a category.

01UH **Definition 1.1.1.1.** The **moduli category of monoidal structures on** C is the category $\mathcal{M}_{\mathbb{E}_1}(C)$ defined by

$$\mathcal{M}_{\mathbb{E}_1}(C) \overset{\text{def}}{=} \mathsf{pt} \times_{\mathsf{Cats}} \mathsf{MonCats}, \qquad \begin{tikzcd} \mathcal{M}_{\mathbb{E}_1}(C) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathsf{MonCats} \arrow[d, "\text{忘}"] \\ \mathsf{pt} \arrow[r, "{[C]}"'] & \mathsf{Cats}. \end{tikzcd}$$

01UJ **Remark 1.1.1.2.** In detail, **the moduli category of monoidal structures on** C is the category $\mathcal{M}_{\mathbb{E}_1}(C)$ where:

- *Objects.* The objects of $\mathcal{M}_{\mathbb{E}_1}(C)$ are monoidal categories $\left(C, \otimes_C, \mathbb{1}_C, \alpha^C, \lambda^C, \rho^C\right)$ whose underlying category is C.
- *Morphisms.* A morphism from $\left(C, \otimes_C, \mathbb{1}_C, \alpha^C, \lambda^C, \rho^C\right)$ to $\left(C, \boxtimes_C, \mathbb{1}'_C, \alpha^{C,\prime}, \lambda^{C,\prime}, \rho^{C,\prime}\right)$ is a monoidal functor structure

$$\begin{aligned} \mathrm{id}^{\otimes}_C &\colon A \otimes_C B \overset{\sim}{\dashrightarrow} A \boxtimes_C B, \\ \mathrm{id}^{\otimes}_{\mathbb{1}|C} &\colon \mathbb{1}_C \overset{\sim}{\dashrightarrow} \mathbb{1}'_C \end{aligned}$$

  on the identity functor $\mathrm{id}_C \colon C \to C$ of C.
- *Identities.* For each $M \overset{\text{def}}{=} \left(C, \otimes_C, \mathbb{1}_C, \alpha^C, \lambda^C, \rho^C\right) \in \mathrm{Obj}\left(\mathcal{M}_{\mathbb{E}_1}(C)\right)$, the unit map

$$\mathbb{1}^{\mathcal{M}_{\mathbb{E}_1}(C)}_{M,M} \colon \mathrm{pt} \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(C)}(M, M)$$

  of $\mathcal{M}_{\mathbb{E}_1}(C)$ at M is defined by

$$\mathrm{id}^{\mathcal{M}_{\mathbb{E}_1}(C)}_M \overset{\text{def}}{=} \left(\mathrm{id}_{\otimes_C}, \mathrm{id}_{\mathbb{1}_C}\right),$$

  where $\left(\mathrm{id}_{\otimes_C}, \mathrm{id}_{\mathbb{1}_C}\right)$ is the identity monoidal functor of **??**.
- *Composition.* For each $M, N, P \in \mathrm{Obj}\left(\mathcal{M}_{\mathbb{E}_1}(C)\right)$, the composition map

$$\circ^{\mathcal{M}_{\mathbb{E}_1}(C)}_{M,N,P} \colon \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(C)}(N, P) \times \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(C)}(M, N) \to \mathrm{Hom}_{\mathcal{M}_{\mathbb{E}_1}(C)}(M, P)$$

  of $\mathcal{M}_{\mathbb{E}_1}(C)$ at (M, N, P) is defined by

$$\left(\mathrm{id}^{\otimes,\prime}_C, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|C}\right) \circ^{\mathcal{M}_{\mathbb{E}_1}(C)}_{M,N,P} \left(\mathrm{id}^{\otimes}_C, \mathrm{id}^{\otimes}_{\mathbb{1}|C}\right) \overset{\text{def}}{=} \left(\mathrm{id}^{\otimes,\prime}_C \circ \mathrm{id}^{\otimes}_C, \mathrm{id}^{\otimes,\prime}_{\mathbb{1}|C} \circ \mathrm{id}^{\otimes}_{\mathbb{1}|C}\right).$$

01UK **Remark 1.1.1.3.** In particular, a morphism in $\mathcal{M}_{\mathbb{E}_1}(C)$ from $\left(C, \otimes_C, \mathbb{1}_C, \alpha^C, \lambda^C, \rho^C\right)$ to $\left(C, \boxtimes_C, \mathbb{1}'_C, \alpha^{C,\prime}, \lambda^{C,\prime}, \rho^{C,\prime}\right)$ satisfies the following conditions:

01UL 1. *Naturality.* For each pair $f\colon A \to X$ and $g\colon B \to Y$ of morphisms of C, the diagram

$$\begin{tikzcd}
A \otimes_C B \arrow[r, "f \otimes_C g"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B}"'] & X \otimes_C Y \arrow[d, "\mathrm{id}^{\otimes}_{X,Y}"] \\
A \boxtimes_C B \arrow[r, "f \boxtimes_C g"'] & X \boxtimes_C Y
\end{tikzcd}$$

commutes.

01UM 2. *Monoidality.* For each $A, B, C \in \mathrm{Obj}(C)$, the diagram

$$\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] & & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] & & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
 & A \boxtimes_C (B \boxtimes_C C) &
\end{tikzcd}$$

commutes.

01UN 3. *Left Monoidal Unity.* For each $A \in \mathrm{Obj}(C)$, the diagram

$$\begin{tikzcd}
 & \mathbb{1}'_C \otimes_C A \arrow[r, "\mathrm{id}^{\otimes}_{\mathbb{1}'_C, A}"] & \mathbb{1}'_C \boxtimes_C A \arrow[dr, "\lambda^{C,\prime}_A"] & \\
\mathbb{1}_C \otimes_C A \arrow[ur, "\mathrm{id}^{\otimes}_{\mathbb{1}} \otimes_C \mathrm{id}_A"] \arrow[rrr, "\lambda^{C}_A"'] & & & A
\end{tikzcd}$$

commutes.

01UP 4. *Right Monoidal Unity.* For each $A \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & A \otimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[r, "\mathrm{id}^{\otimes}_{A,\mathbb{1}'_{\mathcal{C}}}"] & A \boxtimes_{\mathcal{C}} \mathbb{1}'_{\mathcal{C}} \arrow[dr, "\rho^{\mathcal{C},\prime}_{A}"] & \\
A \otimes_{\mathcal{C}} \mathbb{1}_{\mathcal{C}} \arrow[ur, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{\mathbb{1}}"] \arrow[rrr, "\rho^{\mathcal{C}}_{A}"'] & & & A
\end{tikzcd}$$

commutes.

01UQ **Proposition 1.1.1.4.** Let $\mathcal{C}$ be a category.

01UR 1. *Extra Monoidality Conditions.* Let $\left(\mathrm{id}^{\otimes}_{\mathcal{C}}, \mathrm{id}^{\otimes}_{\mathbb{1}|\mathcal{C}}\right)$ be a morphism of $\mathcal{M}_{\mathbb{E}_1}(\mathcal{C})$ from $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ to $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$.

01US (a) The diagram

$$\begin{tikzcd}
(A \otimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \otimes_{\mathcal{C}} \mathrm{id}_C"] \arrow[d, "\mathrm{id}^{\otimes}_{A \otimes_{\mathcal{C}} B, C}"'] & (A \boxtimes_{\mathcal{C}} B) \otimes_{\mathcal{C}} C \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_{\mathcal{C}} B, C}"] \\
(A \otimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C \arrow[r, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_{\mathcal{C}} \mathrm{id}_C"'] & (A \boxtimes_{\mathcal{C}} B) \boxtimes_{\mathcal{C}} C
\end{tikzcd}$$

commutes.

01UT (b) The diagram

$$\begin{tikzcd}
A \otimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \otimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A, B \otimes_{\mathcal{C}} C}"'] & A \otimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C) \arrow[d, "\mathrm{id}^{\otimes}_{A, B \boxtimes_{\mathcal{C}} C}"] \\
A \boxtimes_{\mathcal{C}} (B \otimes_{\mathcal{C}} C) \arrow[r, "\mathrm{id}_A \boxtimes_{\mathcal{C}} \mathrm{id}^{\otimes}_{B,C}"'] & A \boxtimes_{\mathcal{C}} (B \boxtimes_{\mathcal{C}} C)
\end{tikzcd}$$

commutes.

01UU 2. *Mixed Associators.* Let $\left(\mathcal{C}, \otimes_{\mathcal{C}}, \mathbb{1}_{\mathcal{C}}, \alpha^{\mathcal{C}}, \lambda^{\mathcal{C}}, \rho^{\mathcal{C}}\right)$ and $\left(\mathcal{C}, \boxtimes_{\mathcal{C}}, \mathbb{1}'_{\mathcal{C}}, \alpha^{\mathcal{C},\prime}, \lambda^{\mathcal{C},\prime}, \rho^{\mathcal{C},\prime}\right)$ be monoidal structures on $\mathcal{C}$ and let

$$\mathrm{id}^{\otimes}_{-_1,-_2} \colon -_1 \otimes_{\mathcal{C}} -_2 \longrightarrow -_1 \boxtimes_{\mathcal{C}} -_2$$

be a natural transformation.

01UV (a) If there exists a natural transformation

$$\alpha^{\otimes}_{A,B,C}\colon (A \otimes_C B) \boxtimes_C C \to A \otimes_C (B \boxtimes_C C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_C B) \otimes_C C \arrow[r, "\alpha^{C}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A\otimes_C B,C}"'] & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \otimes_C B) \boxtimes_C C \arrow[r, "\alpha^{\otimes}_{A,B,C}"'] & A \otimes_C (B \boxtimes_C C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \otimes_C B) \boxtimes_C C \arrow[r, "\alpha^{\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_C \mathrm{id}_C"'] & A \otimes_C (B \boxtimes_C C) \arrow[d, "\mathrm{id}_{A,B\boxtimes_C C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[r, "\alpha^{C,\prime}_{A,B,C}"'] & A \boxtimes_C (B \boxtimes_C C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

01UW (b) If there exists a natural transformation

$$\alpha^{\boxtimes}_{A,B,C}\colon (A \boxtimes_C B) \otimes_C C \to A \boxtimes_C (B \otimes_C C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_C B) \otimes_C C \arrow[r, "\alpha^{C}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}^{\otimes}_{A,B\otimes_C C}"] \\
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"'] & A \boxtimes_C (B \otimes_C C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A\boxtimes_C B,C}"'] & A \boxtimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \boxtimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[r, "\alpha^{C,\prime}_{A,B,C}"'] & A \boxtimes_C (B \boxtimes_C C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

01UX (c) If there exists a natural transformation

$$\alpha^{\boxtimes,\otimes}_{A,B,C}\colon (A \boxtimes_C B) \otimes_C C \to A \otimes_C (B \boxtimes_C C)$$

making the diagrams

$$\begin{tikzcd}
(A \otimes_C B) \otimes_C C \arrow[r, "\alpha^{C}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"'] & A \otimes_C (B \boxtimes_C C)
\end{tikzcd}$$

and

$$\begin{tikzcd}
(A \boxtimes_C B) \otimes_C C \arrow[r, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] & A \otimes_C (B \boxtimes_C C) \arrow[d, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[r, "\alpha^{C,\prime}_{A,B,C}"'] & A \boxtimes_C (B \boxtimes_C C)
\end{tikzcd}$$

commute, then the natural transformation $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

Proof. *Item 1, Extra Monoidality Conditions*: We claim that Items 1a and 1b are indeed true:

1. *Proof of Item 1a:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms $\mathrm{id}^{\otimes}_{A,B}$ and id_C.

2. *Proof of Item 1b:* This follows from the naturality of $\mathrm{id}^{\otimes}$ with respect to the morphisms id_A and $\mathrm{id}^{\otimes}_{B,C}$.

This finishes the proof.
Item 2, Mixed Associators: We claim that Items 2a to 2c are indeed true:

01UY 1. *Proof of Item 2a :* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of

Item 2 of Remark 1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] \arrow[dd, "\mathrm{id}^{\otimes}_{A \otimes_C B, C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[dd, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] & (1) & A \otimes_C (B \otimes_C C) \arrow[dd, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
 & (A \otimes_C B) \boxtimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \boxtimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{\otimes}_{A,B,C}"] & (2) \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] & (3) & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
 & A \boxtimes_C (B \boxtimes_C C) &
\end{tikzcd}$$

Since:

- Subdiagram (1) commutes by Item 1a of Item 1.
- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by assumption.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

01UZ 2. *Proof of Item 2b :* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of

Item 2 of Remark 1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[dd, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] \arrow[dr, "\alpha^{\boxtimes}_{A,B,C}"] \arrow[rr, phantom, "(1)"] & & A \otimes_C (B \otimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \otimes_C C}"] \arrow[dd, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
 & A \boxtimes_C (B \otimes_C C) \arrow[dd, "\mathrm{id}_A \boxtimes_C \mathrm{id}^{\otimes}_{B,C}"] & \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] \arrow[rr, phantom, "(2)"] & & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \arrow[rr, phantom, "(3)"] & \\
 & A \boxtimes_C (B \boxtimes_C C) &
\end{tikzcd}$$

Since:

- Subdiagram (1) commutes by assumption.
- Subdiagram (2) commutes by assumption.
- Subdiagram (3) commutes by Item 1b of Item 1.

it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

01V0 3. *Proof of Item 2c :* We may partition the monoidality diagram for $\mathrm{id}^{\otimes}$ of

Item 2 of Remark 1.1.1.3 as follows:

$$\begin{tikzcd}
 & (A \otimes_C B) \otimes_C C \arrow[dl, "\mathrm{id}^{\otimes}_{A,B} \otimes_C \mathrm{id}_C"'] \arrow[dr, "\alpha^{C}_{A,B,C}"] & \\
(A \boxtimes_C B) \otimes_C C \arrow[d, "\mathrm{id}^{\otimes}_{A \boxtimes_C B, C}"'] \arrow[drr, "\alpha^{\boxtimes,\otimes}_{A,B,C}"] \arrow[rr, phantom, "(1)"] & & A \otimes_C (B \otimes_C C) \arrow[d, "\mathrm{id}_A \otimes_C \mathrm{id}^{\otimes}_{B,C}"] \\
(A \boxtimes_C B) \boxtimes_C C \arrow[dr, "\alpha^{C,\prime}_{A,B,C}"'] \arrow[rr, phantom, "(2)"] & & A \otimes_C (B \boxtimes_C C) \arrow[dl, "\mathrm{id}^{\otimes}_{A, B \boxtimes_C C}"] \\
 & A \boxtimes_C (B \boxtimes_C C) &
\end{tikzcd}$$

Since subdiagrams (1) and (2) commute by assumption, it follows that the boundary diagram also commutes, i.e. $\mathrm{id}^{\otimes}$ satisfies the monoidality condition of Item 2 of Remark 1.1.1.3.

This finishes the proof. □

1.2 The Moduli Category of Braided Monoidal Structures on a Category

01V1

1.3 The Moduli Category of Symmetric Monoidal Structures on a Category

01V2

01V3 2 Moduli Categories of Closed Monoidal Structures

3 Moduli Categories of Refinements of Monoidal Structures

01V4

3.1 The Moduli Category of Braided Refinements of a Monoidal Structure

01V5

Appendices

A Other Chapters

Sets

1. Sets
2. Constructions With Sets
3. Monoidal Structures on the Category of Sets
4. Pointed Sets
5. Tensor Products of Pointed Sets

Relations

6. Relations
7. Constructions With Relations
8. Equivalence Relations and Apartness Relations

Category Theory

9. Categories

Monoidal Categories

10. Constructions With Monoidal Categories

Bicategories

11. Types of Morphisms in Bicategories